

VENDOR PERFORMANCE ANALYSIS

PROJECT OVERVIEW

This project analyzes vendor performance using procurement, sales, inventory, and freight datasets to uncover insights into profitability, procurement efficiency, vendor contribution, and operational costs. The objective was to help businesses optimize vendor relationships, improve inventory planning, and increase profitability through data-driven decision-making.

Dataset Summary

- Multiple transactional datasets used
- Key tables included:
 - Purchases
 - Purchase Prices
 - Vendor Invoices
 - Sales
 - Vendor Sales Summary

Key Features

- Vendor information
 - Product and brand details
 - Purchase quantities and costs
 - Sales revenue
 - Gross profit
 - Freight costs
 - Inventory metrics
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Data Preparation and Cleaning using Python

The project started with data preparation and ingestion using Python.

Data Loading

- Imported datasets using pandas.
- Connected Python with SQLite database.
- Loaded CSV files into structured database tables.

Data Cleaning

- Checked for missing values.
- Standardized column names.
- Verified data consistency.
- Removed redundant and inconsistent records.

ETL Pipeline

- Implemented chunk-based ingestion for large datasets.
- Automated CSV-to-database loading.
- Used logging mechanism for monitoring ingestion status.

Feature Engineering

Created business metrics such as:

- Gross Profit
 - Profit Margin
 - Purchase-to-Sales Ratio
 - Vendor Efficiency Metrics
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Exploratory Data Analysis using Python

EDA was performed to identify business trends, vendor behavior, and profitability patterns.

Analysis Performed

Descriptive Statistics

- Used mean, median, standard deviation, and quartiles.
- Analyzed revenue and profit distributions.

Distribution Analysis

Created histograms for:

- Sales Revenue
- Purchase Amount
- Gross Profit
- Freight Costs

Outlier Detection

Used boxplots to identify:

- High-volume vendors
- Unusual freight costs
- High-margin brands

Correlation Analysis

Generated heatmaps to study relationships between:

- Purchase Dollars
- Sales Revenue
- Gross Profit
- Inventory Metrics

Key Insight

Strong positive relationships were observed between purchase volume, sales revenue, and profitability.

Data Analysis using SQL

SQL queries were used extensively to answer business questions and generate analytical insights.

Business Analysis Performed

1. Vendor Revenue Analysis
 - Identified top revenue-generating vendors.
 2. Profitability Analysis
 - Calculated gross profit and profit margins vendor-wise.
 3. Freight Cost Analysis
 - Evaluated vendors with high operational and freight costs.
 4. Brand Performance Analysis
 - Identified high-margin brands and low-performing products.
 5. Procurement Efficiency
 - Analyzed Purchase-to-Sales ratios.
 6. Vendor Segmentation
 - Classified vendors based on revenue contribution and profitability.
 7. Inventory Analysis
 - Studied purchasing trends and inventory movement.
 8. Sales Contribution Analysis
 - Evaluated contribution of vendors and brands toward total sales.
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Statistical Analysis

The project used statistical methods to improve business understanding.

Techniques Used

- Correlation Analysis
- Quantile Analysis
- Distribution Analysis
- Vendor Segmentation
- Outlier Detection

Key Findings

- A small number of vendors contributed disproportionately to revenue.
- Several low-volume brands generated high profit margins.
- Freight costs varied significantly across vendors.

Dashboard in Power BI

An interactive Power BI dashboard can be created using the analytical dataset.

Recommended Dashboard KPIs

- Total Sales Revenue
- Gross Profit
- Profit Margin %
- Freight Cost
- Vendor Contribution
- Top Brands
- Purchase-to-Sales Ratio
- Inventory Efficiency

Recommended Dashboard Pages

1. Executive Overview
 2. Vendor Performance
 3. Profitability Insights
 4. Brand Analysis
 5. Freight Cost Analysis
 6. Procurement Efficiency
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Key Business Insights

Revenue Concentration

A small percentage of vendors generated a major portion of overall revenue.

Hidden Profit Opportunities

Several brands showed high profit margins despite low sales volume.

Procurement-Sales Relationship

Purchase quantities strongly influenced sales performance.

Freight Optimization Opportunity

Certain vendors had disproportionately high freight costs.

Inventory Efficiency Differences

Vendor efficiency varied significantly based on Purchase-to-Sales ratios.

Business Recommendations

Optimize Vendor Management

Focus more on high-performing and profitable vendors.

Improve Freight Efficiency

Negotiate logistics and freight costs with expensive vendors.

Promote High-Margin Brands

Increase visibility and inventory allocation for high-margin products.

Improve Procurement Planning

Use sales trends to reduce over-purchasing and improve inventory turnover.

Implement Real-Time Monitoring

Deploy Power BI dashboards for continuous performance tracking.

Skills Demonstrated

Technical Skills

- Python
- SQL
- SQLite
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Data Cleaning
- Exploratory Data Analysis
- ETL Pipeline Development
- Statistical Analysis

Business Analytics Skills

- Vendor Analysis
- Profitability Analysis
- Procurement Analytics
- KPI Design
- Business Intelligence

- Insight Generation
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Conclusion

This project successfully transformed raw transactional vendor data into actionable business insights using Python, SQL, and analytical techniques.

The analysis identified:

- High-performing vendors
- Procurement inefficiencies
- Freight optimization opportunities
- Profit-driving brands
- Vendor profitability trends